

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1200

Refuse 69 kV
August 24, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1200. The Interconnection Customer proposes a 93.5 MW battery energy storage facility interconnecting at the Refuse 69 kV substation in the North Carolina area. The request was received on May 02, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 410 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,500,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W SACKELFORD – N FOND DU LAC 138 KV	TECUMSEH ENERGY CENTER – GASTON 345 KV Line	103.4	225	5.1
TROWBRIDGE 138-69 kV XFMR Bank 1	HOME – BEL AIR 138 KV	102.8	92	27.0
W SHAKELFORD – RALEIGH HILLS 138 KV	TECUMSEH ENERGY CTR – GASTON 345 KV	101.3	218	5.1
RALEIGH HILLS – EAGLE ROCK 138 KV line	GASTON – TECUMSEH ENERGY CTR 345 KV line	100.5	208	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W SHACKELFORD – NORTH FOND DU LAC 138 KV	Reconductor line	\$3,754,000	24% (\$921,000)
TROWBRIDGE 138-69 kV XFMR Bank 1	Replace transformer with higher-rated unit	\$4,704,000	29% (\$1,384,000)
W SHACKELFORD – RALEIGH HILLS 138 KV	Reconductor line	\$3,101,000	24% (\$760,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RALEIGH HILLS – EAGLE ROCK 138 KV line	Reconductor line	\$1,775,000	24% (\$435,000)
	Total allocated to project		\$3,500,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.